CFRM 502, Winter 2020.

Homework 4

Due: Monday Febraury 17th, 2020, NO LATER than 11:59pm

Instructions: You must submit your work through Canvas, and in the form of two attachments: A .pdf file of your detailed answers and a file containing your codes (.R or .Rmd). The .pdf files can be typeset or be a readable scan of your handwritten notes. You do not need to include R outputs in the .pdf file. However, you must clearly indicate the relevant part of your R codes when answering a question that is based on your code.

Grading: 3 questions (out of 5) will be graded and contribute to 80% of the score of this Homework. The remaining 20% is awarded if genuine effort is spend on answering the remaining questions.

1. Consider the daily VIX index of the Chicago Board Options Exchange (CBOE) from January 2, 1990 to March 31, 2015. You may obtain this data (and remove missing values) by running the following R code.

```
getSymbols("VIXCLS", src="FRED")
VIX = na.omit(VIXCLS["1990-01/2015-03"])
```

   Alternatively, you may load that data from the text file "HW4-Q1.txt" that is posted with this homework, using

```
library(xts)
dat=read.table("HW4-Q1.txt", header=TRUE, sep = ",")
VIX=xts(dat[,-1], order.by=as.Date(dat[, 1]))
names(VIX) = c("VIXCLS")
```

   The resulting variable "VIX" should have 6359 observations. Let $v_t = \log(\text{VIX}_t)$.

   (a) Instead of using a proper time series model, we decide to use a simple line regression model

$$v_t = \beta_0 + \beta_1 v_{t-1} + \varepsilon_t$$

   Fit this model (i.e. express the estimated values $\widehat{\beta}_0$ and $\widehat{\beta}_1$). Provide a scatter plot of v_t versus its lagged value v_{t-1} and include the fitted line.

   (b) Do a full regression diagnostic on the fitted model in part (a).

   (c) Is the model in part (a) appropriate for modeling the VIX time series? In other words, can it be used for forecasting future VIX values? Explain your answer.

2. Consider the VIX growth rate series $z_t = v_t - v_{t-1}$, where v_t is as in Problem 1.

   (a) Does VIX have zero growth rate? That is, test for $H_0 : \mathbb{E}(z_t) = 0$. Draw your conclusion.

   (b) Is $\{z_t\}$ autocorrelated? Support your answer with appropriate tests.

   (c) Fit an AR(1) model to $\{z_t\}$ and state the estimated model. Is this model a good fit to $\{z_t\}$? Explain your answer and provide supporting graphs and/or tests.

3. (a) Is the AR(4) model

$$r_t = 20 - \frac{1}{90} r_{t-1} + \frac{1}{30} r_{t-2} + \frac{19}{90} r_{t-3} - \frac{3}{10} r_{t-4} + \varepsilon_t,$$

weakly stationary? Justify your answer. Hint: You may use the R function "polyroot" to find roots of polynomial functions, and "Mod" to obtain the absolute value of complex numbers.

(b) Assume that the AR(p) model $r_t = \phi_0 + \sum_{h=1}^{p} \phi_h r_{t-h} + \varepsilon_t$ is weakly stationary. Find $\mathbb{E}(r_t)$ for all t.

(c) Consider the AR(2) model $r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \varepsilon_t$ where $\{\varepsilon_t\}$ is weak WN$(0, \sigma_\varepsilon^2)$. Assume that $\{r_t\}$ is weakly stationary. Calculate $\mathrm{Var}(r_t)$ and $\mathrm{Cov}(r_t, r_{t-1})$ for all t.

4. Consider the monthly crude oil prices from January 1987 to January 2018. Let $\{P_t\}$ be the monthly oil prices, and $r_t = \Delta P_t$ be the monthly price changes. The data can be obtained from FRED via "quantmod". Alternatively, you may use the following code to obtain the data from the text file "HW4-Q4.txt" that is posted with this homework.

```
library(xts)
dat=read.table("HW4-Q4.txt", header=TRUE, sep = ",")
P=xts(dat[,2], order.by=as.Date(dat[, 1]));names(P) = c("Oil")
r = diff(P)[-1]
```

   (a) Is $\{P_t\}$ a weak white noise? How about $\{r_t\}$? Run relevant tests, state their hypotheses and draw conclusions.

   (b) Fit an AR model for $\{r_t\}$ and state the fitted model. To choose the AR order, use the sample PACF plot. Is your model a good specification of the data? Support your answer with appropriate plots and tests.

   (c) Fit an MA model to $\{r_t\}$. State the fitted model. Use the sample ACF plot to decide on MA order. Is your MA model adequate?

   (d) Use information criteria AIC and BIC to choose an ARIMA(p,d,q) for $\{P_t\}$. Comment on adequacy of your model.

5. Use the following code to obtain global temperatures from the text file "HW4-Q5.txt" that accompanies this homework.

```
dat = read.table("HW4-Q5.txt",header=T)
dd = dat[,2:13]
x  = c(t(dd))
```

   The data includes temperature measurements from January 1880 to December 2015 for the Northern Hemisphere, in 0.01C. Let $\{x_t\}$ be the global temperatures.

   (a) Is the temperature series a stationary process? Conduct three relevant test, state their hypotheses and conclusions.

   (b) Let $z_t = x_t - x_{t-1}$ be the monthly change in the global temperature. Is the average monthly changes in the temperature zero? Is the process $\{z_t\}$ stationary? State appropriate statistical tests, and draw your conclusion.

   (c) Fit an ARIMA model to $\{x_t\}$. State the fitted model and conduct a full model diagnostic.